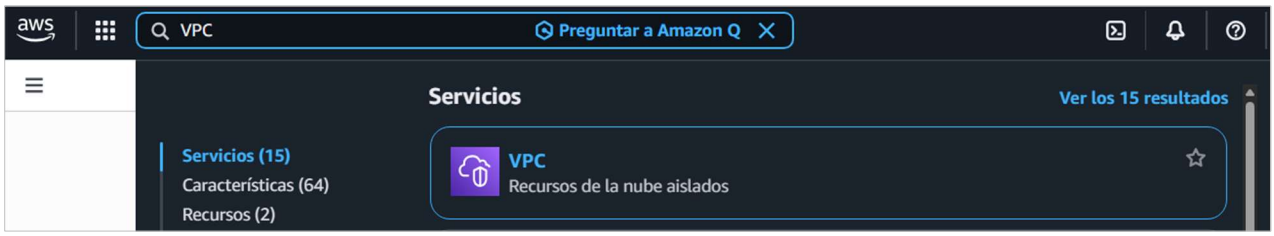
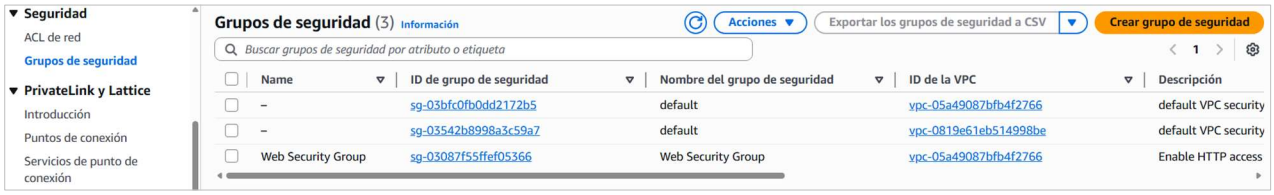
162--Lab - [Reto] Crear y acceder a un servidor RDS

En esta práctica se muestra como lanzar una instancia de Amazon Aurora y ejecutar comandos SQL en la base de datos desde la terminal de Linux de una instancia EC2.

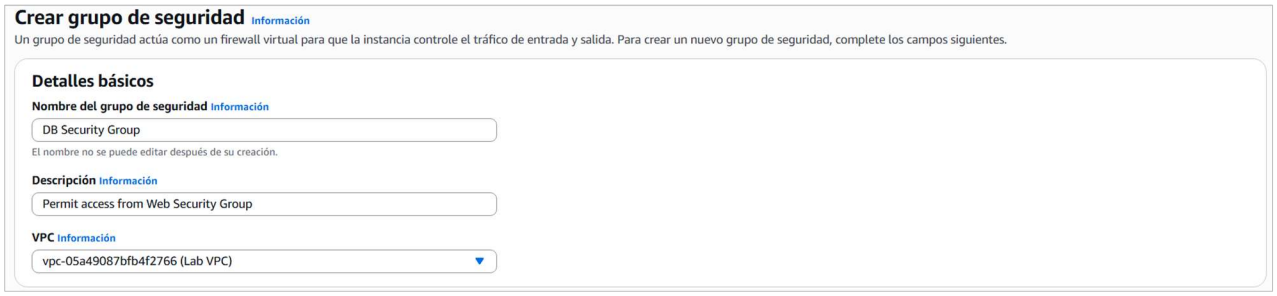
Crear un grupo de seguridad para la base de datos

Primero necesitamos crear el grupo de seguridad que asignaremos a la instancia de Amazon Aurora

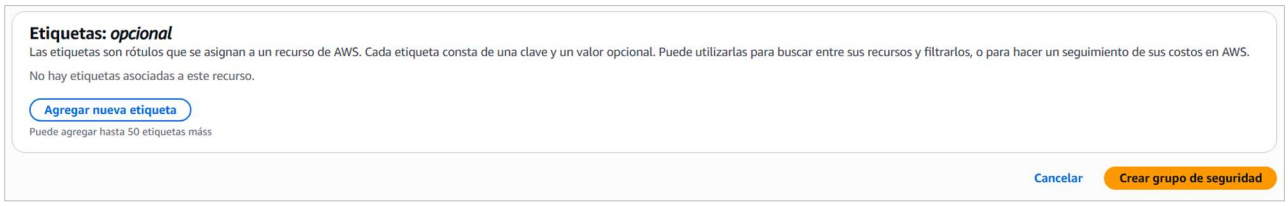




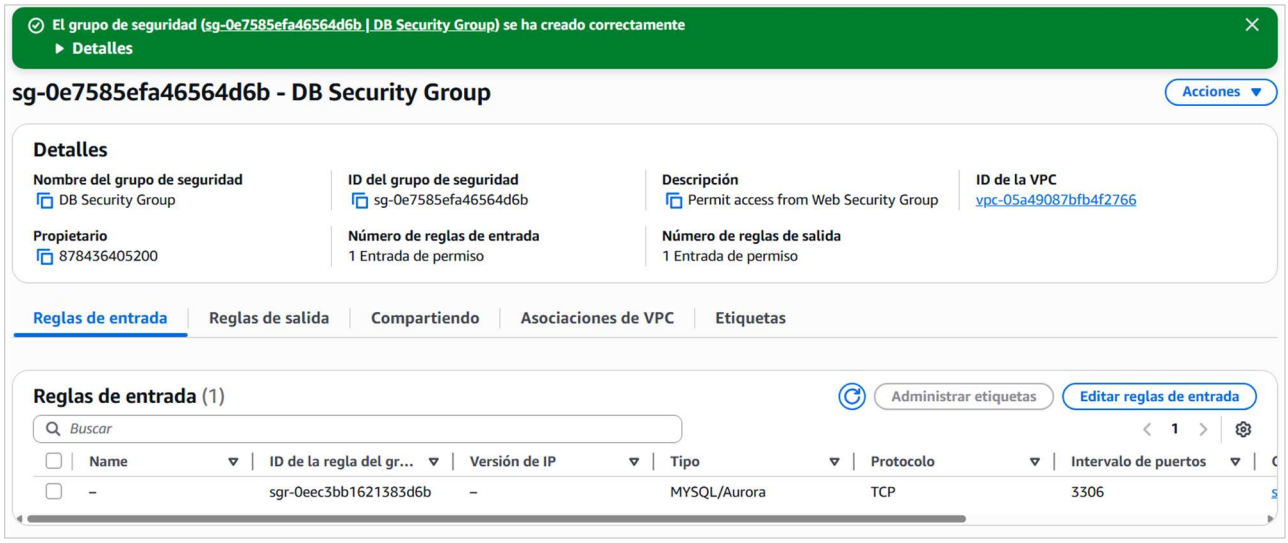
El nombre del grupo de seguridad es "Db Security Group"



Indicamos que el grupo de seguridad permitirá la conexión desde el grupo de seguridad Web Security Group, que es el grupo al que pertenece la instancia EC2 desde la que nos conectaremos a la base de datos

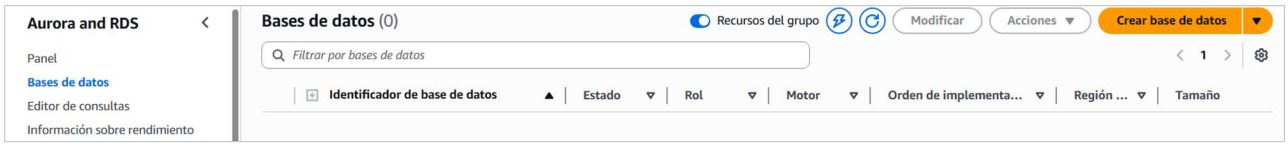


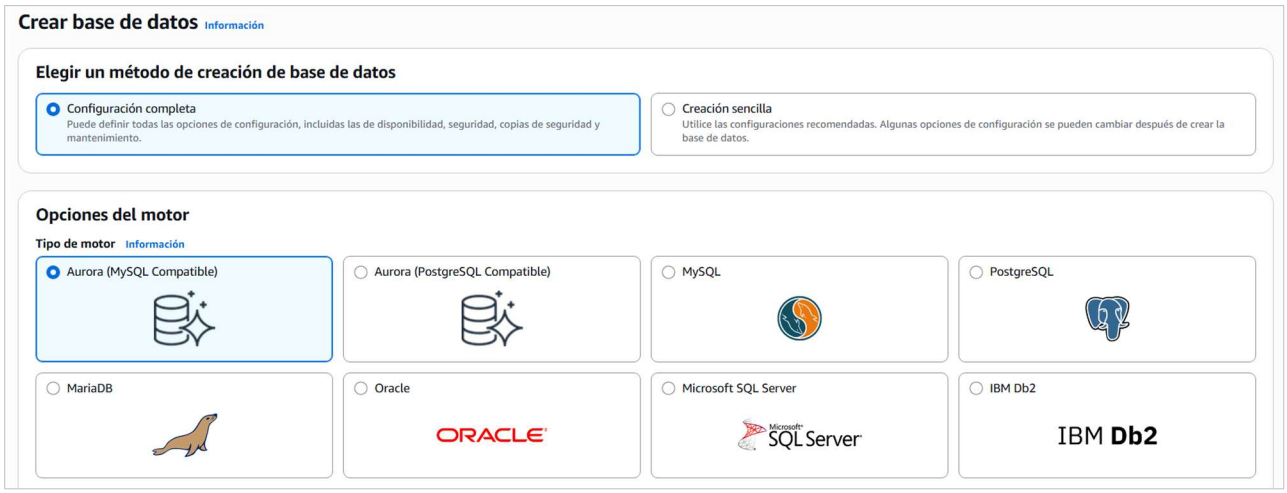
El grupo de seguridad ha sido creado

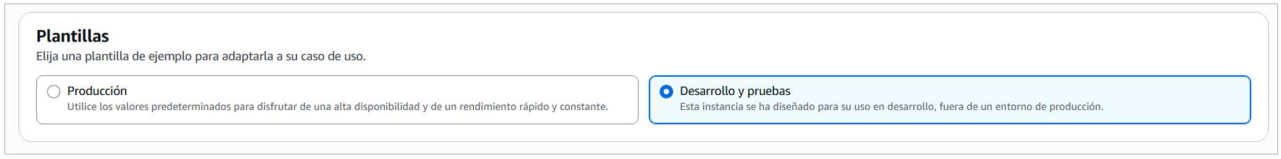


Lanzar una instancia de Amazon Aurora

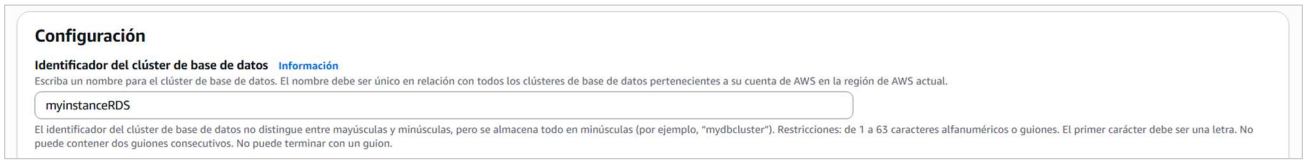
Ahora lanzaremos la instancia de Amazon Aurora



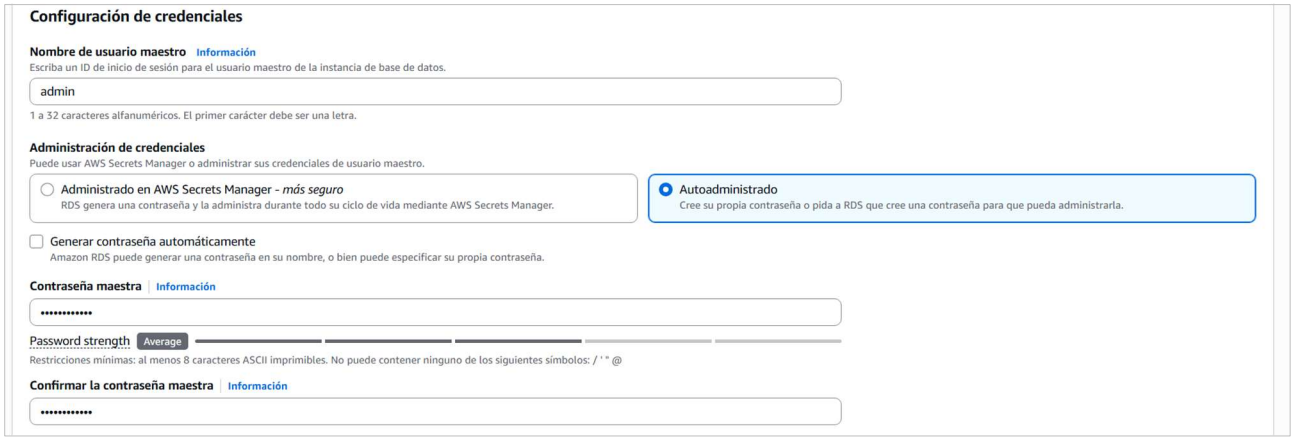


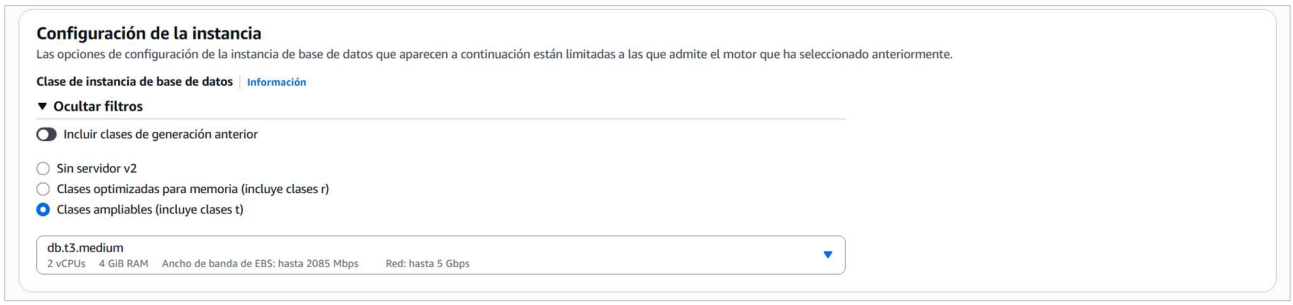


Definimos el nombre de la instancia

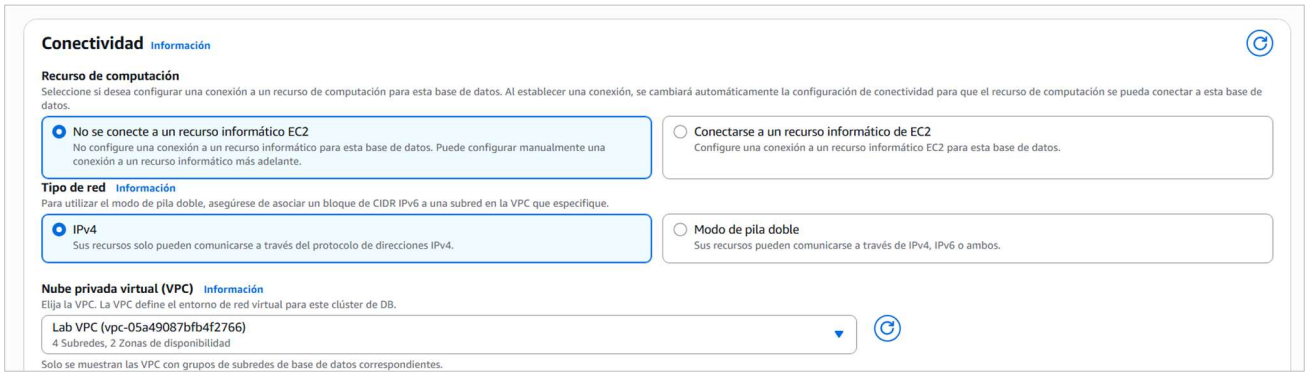


Definimos el usuario y la contraseña de acceso

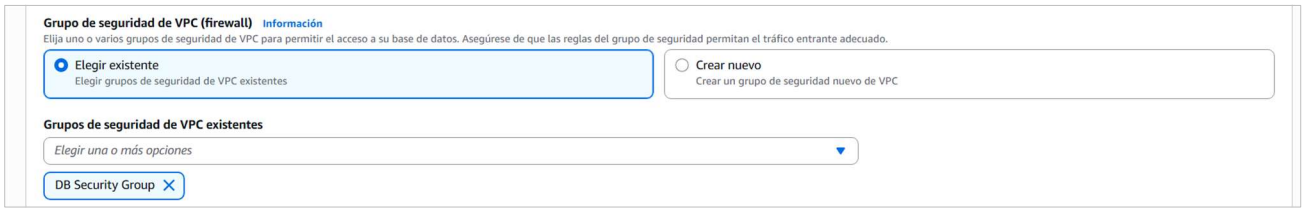


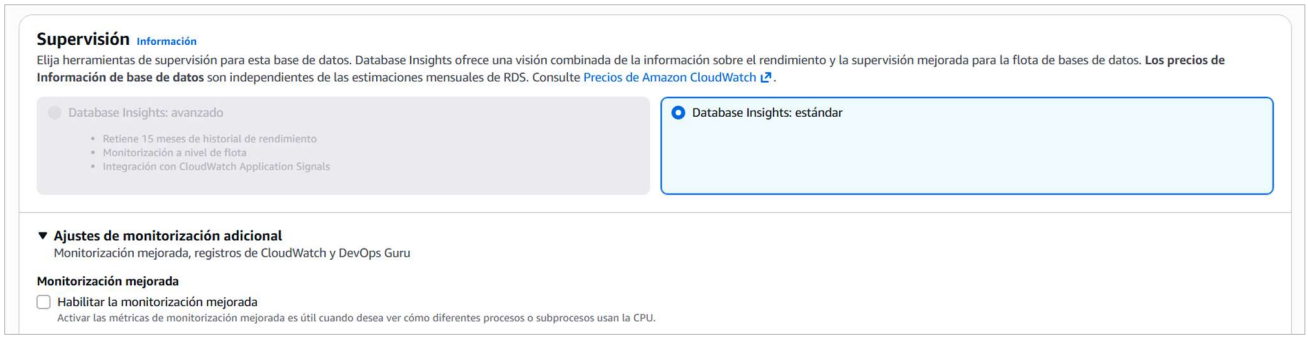


Seleccionamos la VPC "Lab VPC"

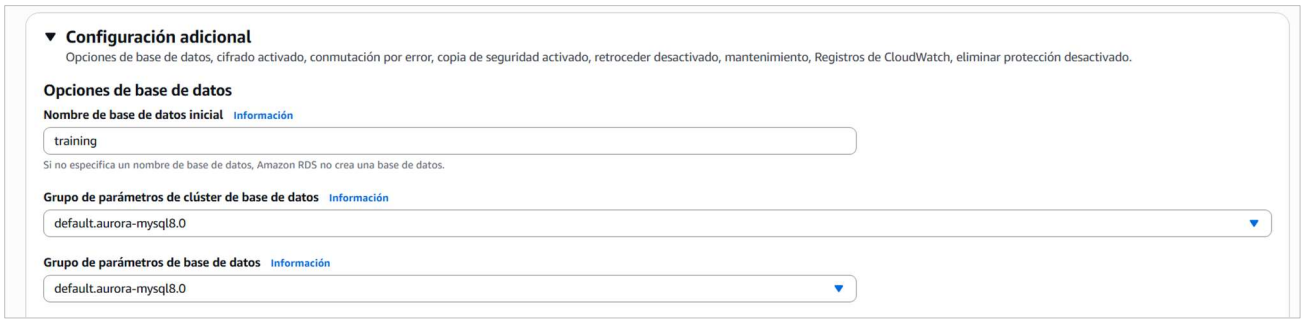


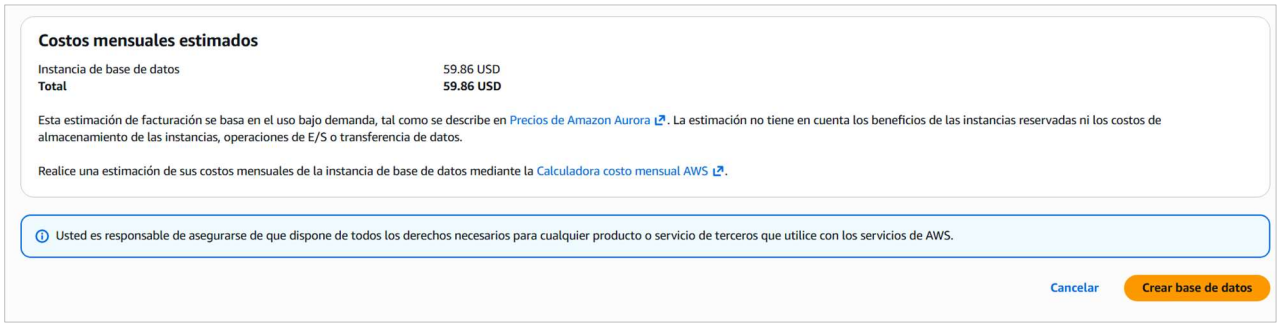
Seleccionamos el grupo de seguridad que creamos anteriormente



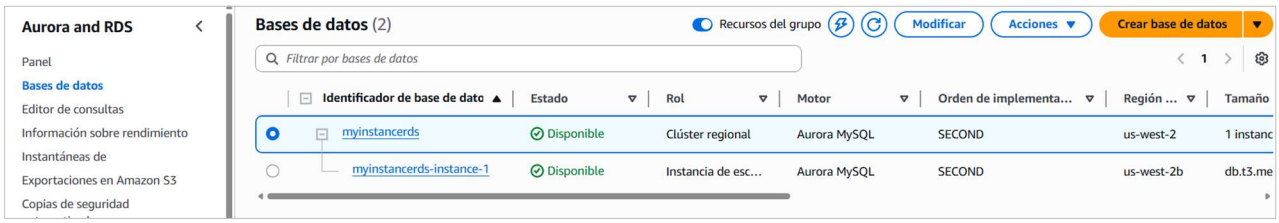


Especificamos el nombre de la base de datos que creará al lanzar la instancia



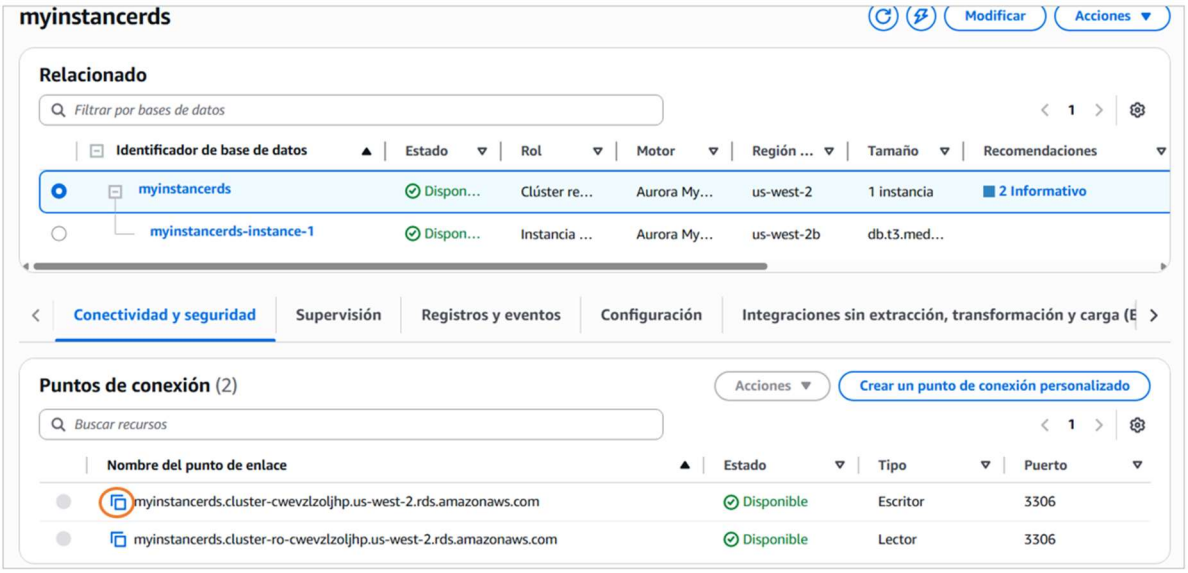


La base de datos ha sido creada



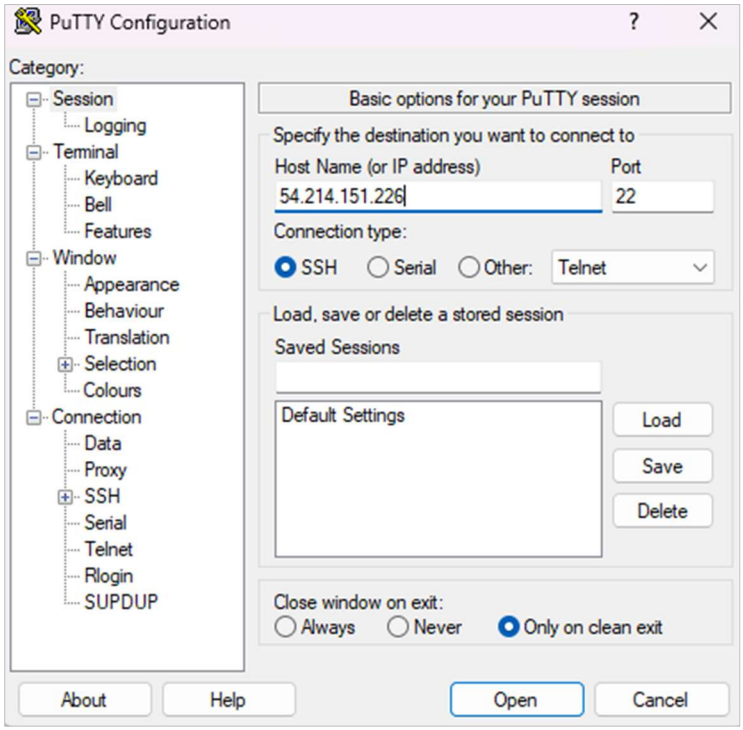
Copiamos el endpoint de la instancia de escritura:

myinstancerds.cluster-cwevzlzoljhp.us-west-2.rds.amazonaws.com



Conectar a la instancia EC2 y ejecutar comandos SQL

Ahora nos conectamos a la instancia EC2 por medio del protocolo SSH y la dirección IPv4 pública de la instancia



Instalamos el cliente de MariaDB para poder interactuar con la base de datos

```
[ec2-user@ip-10-0-2-165 ~]$ sudo yum install mariadb
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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Resolving Dependencies
--> Running transaction check
---> Package mariadb.x86_64 1:5.5.68-1.amzn2.0.1 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=================================================================================================================================================================================================================================================================================================================================================================================================================================================================================================
 Package                        Arch                        Version                                    Repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
Installing:
 mariadb                        x86_64                      1:5.5.68-1.amzn2.0.1                       amzn2-core                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
Transaction Summary
=================================================================================================================================================================================================================================================================================================================================================================================================================================================================================================
Install  1 Package

Total download size: 8.8 M
Installed size: 49 M
Is this ok [y/d/N]: clear
Is this ok [y/d/N]: y
Downloading packages:
mariadb-5.5.68-1.amzn2.0.1.x86_64.rpm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  Verifying  : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                

Installed:
  mariadb.x86_64 1:5.5.68-1.amzn2.0.1

Complete!
```

Ahora nos conectamos a la base de datos utilizando el usuario, la contraseña y el endpoint de la instancia de escritura:

mysql -u admin --password='lab-password' -h myinstancerds.cluster-cwevzlzoljhp.us-west-2.rds.amazonaws.com

```
[ec2-user@ip-10-0-2-165 ~]$ mysql -u admin --password='lab-password' -h  myinstancerds.cluster-cwevzlzoljhp.us-west-2.rds.amazonaws.com
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 52
Server version: 8.0.42 6252a59a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

Mostramos las bases de datos disponibles

```
MySQL [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| training           |
+--------------------+
5 rows in set (0.00 sec)
```

En la base de datos *training* creamos la tabla *RESTART*

```
MySQL [(none)]> USE training;
Database changed
MySQL [training]> CREATE TABLE RESTART  ( StudentID INT NOT NULL, StudentName VARCHAR(100) NOT NULL, RestartCity VARCHAR(100), Grad
uationDate DATETIME, PRIMARY KEY (StudentID) );
Query OK, 0 rows affected (0.02 sec)
```

Insertamos datos en la tabla *RESTART*

```
MySQL [(none)]> USE training;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [training]> INSERT INTO RESTART  (StudentID, StudentName, RestartCity, GraduationDate)
    -> VALUES (1, 'Carolina Bolayna', 'Guadalajara', '2025-12-15 10:00:00'),
    -> (2, 'Cecilia Pérez', 'Lima', '2025-12-15 11:30:00'),
    -> (3, 'Roberto Méndez', 'Ciudad de México', '2026-01-10 09:00:00'),
    -> (4, 'Luis Torres', 'Lima', '2026-01-12 15:45:00'),
    -> (5, 'Mariana Ramos', 'Michoacán', '2026-01-15 08:20:00'),
    -> (6, 'David Huitrón', 'Villahermosa', '2025-11-20 14:00:00'),
    -> (7, 'Julio Jimenez', 'Nuevo Laredo', '2025-12-20 10:30:00'),
    -> (8, 'Karla Savala', 'Puerto Vallarta', '2026-02-05 17:00:00'),
    -> (9, 'María Inés Fernandez', 'Monterrey', '2026-03-01 12:00:00'),
    -> (10, 'Javier Montiel', 'Oaxaca', '2025-10-30 09:15:00');
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Consultamos la tabla *RESTART*

```
MySQL [training]> SELECT * FROM RESTART;
+-----------+-----------------------+-------------------+---------------------+
| StudentID | StudentName           | RestartCity       | GraduationDate      |
+-----------+-----------------------+-------------------+---------------------+
|         1 | Carolina Bolayna      | Guadalajara       | 2025-12-15 10:00:00 |
|         2 | Cecilia Pérez         | Lima              | 2025-12-15 11:30:00 |
|         3 | Roberto Méndez        | Ciudad de México  | 2026-01-10 09:00:00 |
|         4 | Luis Torres           | Lima              | 2026-01-12 15:45:00 |
|         5 | Mariana Ramos         | Michoacán         | 2026-01-15 08:20:00 |
|         6 | David Huitrón         | Villahermosa      | 2025-11-20 14:00:00 |
|         7 | Julio Jimenez         | Nuevo Laredo      | 2025-12-20 10:30:00 |
|         8 | Karla Savala          | Puerto Vallarta   | 2026-02-05 17:00:00 |
|         9 | María Inés Fernandez  | Monterrey         | 2026-03-01 12:00:00 |
|        10 | Javier Montiel        | Oaxaca            | 2025-10-30 09:15:00 |
+-----------+-----------------------+-------------------+---------------------+
10 rows in set (0.00 sec)
```

Creamos la tabla *CLOUD_PRACTITIONER,* con una llave foránea a la *tabla RESTART*

```
MySQL [training]> CREATE TABLE CLOUD_PRACTITIONER ( ID INT NOT NULL, CertificationDate DATETIME, StudentID INT, PRIMARY KEY (ID), FOREIGN K
EY (StudentID) REFERENCES RESTART(StudentID) );
Query OK, 0 rows affected (0.03 sec)
```

Insertamos datos en la tabla *CLOUD_PRACTITIONER*

```
MySQL [training]> INSERT INTO CLOUD_PRACTITIONER (ID, CertificationDate, StudentID) VALUES (1, '2025-11-25 09:00:00', 1), (2, '2025
-11-28 14:30:00', 4), (3, '2025-12-05 10:15:00', 5), (4, '2025-12-10 16:45:00', 8), (5, '2025-12-12 11:00:00', 10);
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

Consultamos la tabla *CLOUD_PRACTITIONER*

```
MySQL [training]> SELECT * FROM CLOUD_PRACTITIONER;
+----+---------------------+-----------+
| ID | CertificationDate   | StudentID |
+----+---------------------+-----------+
|  1 | 2025-11-25 09:00:00 |         1 |
|  2 | 2025-11-28 14:30:00 |         4 |
|  3 | 2025-12-05 10:15:00 |         5 |
|  4 | 2025-12-10 16:45:00 |         8 |
|  5 | 2025-12-12 11:00:00 |        10 |
+----+---------------------+-----------+
5 rows in set (0.00 sec)
```

Hacemos un inner join entre las tablas *RESTART* y *CLOUD_PRACTITIONER*

```
MySQL [training]> SELECT
    ->     RESTART.StudentID,
    ->     RESTART.StudentName,
    ->     CLOUD_PRACTITIONER.CertificationDate
    -> FROM
    ->     RESTART
    -> INNER JOIN
    ->     CLOUD_PRACTITIONER
    -> ON
    ->     RESTART.StudentID = CLOUD_PRACTITIONER.StudentID;
+-----------+-----------------+---------------------+
| StudentID | StudentName     | CertificationDate   |
+-----------+-----------------+---------------------+
|         1 | Carolina Bolayna | 2025-11-25 09:00:00 |
|         4 | Luis Torres     | 2025-11-28 14:30:00 |
|         5 | Mariana Ramos   | 2025-12-05 10:15:00 |
|         8 | Karla Savala    | 2025-12-10 16:45:00 |
|        10 | Javier Montiel  | 2025-12-12 11:00:00 |
+-----------+-----------------+---------------------+
5 rows in set (0.00 sec)
```